

Table 2: Estimation of AR(1) with Different Moments

Moments	SMM ($R = \infty$)		NNE ($L = 1e3$)	
	Bias	RMSE	Bias	RMSE
1) $y_i y_{i-1}$	-0.019 (.003)	0.093 (.002)	-0.017 (.003)	0.091 (.002)
2) $y_i y_{i-1}, y_i^2$	-0.015 (.003)	0.088 (.002)	-0.013 (.003)	0.086 (.002)
3) $y_i y_{i-k}, k = 1, 2, 3$	-0.026 (.003)	0.099 (.003)	-0.015 (.003)	0.091 (.002)
4) $y_i y_{i-k}, k = 1, 2, \dots, 10$	-0.043 (.003)	0.111 (.003)	-0.014 (.003)	0.095 (.002)
5) $y_i y_{i-1}, y_i^2 y_{i-1}, y_i y_{i-1}^2$	-0.075 (.003)	0.130 (.003)	-0.014 (.003)	0.095 (.002)
6) $y_i y_{i-k}, y_i^2 y_{i-k}, y_i y_{i-k}^2, k = 1, 2, 3$	-0.086 (.003)	0.135 (.003)	-0.013 (.003)	0.096 (.002)

Notes: SMM ($R = \infty$) is the same as GMM and follows the usual 2-step procedure. The 1st step constructs optimal weighting matrix accounting for serial correlations. NNE ($L = 1e3$) uses a shallow neural net of 32 hidden nodes. Reported numbers are based on 1000 Monte Carlo datasets. Numbers in parentheses are standard errors.

$\hat{\mathbf{f}}(\beta) = \mathbf{m}$ typically has no exact solution.

To summarize, Figure 2 shows that SMM and NNE take different approaches and subsequently incur different types of estimation errors. While both rely on approximations of the mapping between moment and parameter, SMM makes a *point-by-point approximation from parameter to moment* whereas NNE makes a *functional approximation from moment to parameter*. It is not difficult to see that in cases where the moment evaluation has sizable simulation errors (so that the point-by-point approximation is noisy), the functional approach by NNE can have advantages. A similar comparison applies to NNE and indirect inference (see Appendix A.6).

3.3 Robustness to redundant moments

So far, we have focused on the estimation relying a single moment for the purpose of visual illustration. Next, we move to estimation using multiple moments to examine NNE’s robustness to redundant moments in the context of the AR(1) model. As discussed in Section 2, it is known that redundant moments can introduce substantial biases to SMM in finite samples. In comparison, NNE should be less affected by redundant moments thanks to its learning mechanism.

We use Monte Carlo experiments to compare SMM and NNE with multiple moments. The simplicity of AR(1) allows us to choose $R = \infty$ for SMM, which is just GMM. This choice removes simulation errors and thus helps to isolate the impact of redundant moments. For NNE, we use $L = 1000$. Our Monte Carlo exercise sets $\beta = 0.6$ and generates 1000 datasets using the AR(1) model. We estimate β from each of these Monte Carlo datasets.

Table 2 shows the results under six different moment specifications. Row 1 is our benchmark relying on the single moment $\frac{1}{n-1} \sum_{i=2}^n y_i y_{i-1}$. Row 2 adds the moment $\frac{1}{n} \sum_{i=1}^n y_i^2$. Note that $\mathbb{E}(y_i^2) = 1/(1 - \beta^2)$ so the moment is informative about β . We see a slight reduction in the RMSE for GMM, suggesting that in this case the information gain from the added moment overcomes the potential bias increase. We also see a slight reduction in the RMSE for NNE, showing that NNE is able to capitalize on the information from the added moment too.

Row 3 and 4 add moments involving further lags to the benchmark case. Note that $\mathbb{E}(y_i y_{i-k}) =$